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EDITED BY
Galina Ilieva,
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Sandeep Singh Sengar,
Cardiff Metropolitan University,
United Kingdom
Noble Lo,
Lancaster University, United Kingdom

*CORRESPONDENCE
Kai Cao
caokai19910101@gmail.com

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The moderating role of ethnic culture on adoption intention of generative artificial intelligence among university students

Kai Cao[1]*, Ping Wang [2,3] and Jie Zhao[4]

[1]Library of Qinghai University, Qinghai University, Xining, China, [2]Institute of Education, Xiamen University, Xiamen, China, [3]School of International Education, Qinghai Minzu University, Xining, China, [4]Institute of Mental Health Education, Jining University, Jining, China

This study is based on an extended model of the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) to explore the influencing factors of the adoption intention of generative artificial intelligence (GAI) in the higher-education environment of multi-ethnic regions. It also focuses on analyzing the moderating effects of ethnic culture. Data from 432 university students were collected through a questionnaire survey and analyzed using a structural equation model. The research findings are as follows: (1) The factors influencing university students' adoption intention of generative artificial intelligence include perceived ease of use, perceived usefulness, social influence, facilitating conditions, perceived learning performance, perceived learning efficiency, and user satisfaction. (2) Tool designs that are not well-adapted to ethnic cultures and the conflicts between practicality and cultural values weaken users' perceived usefulness of generative artificial intelligence, negatively moderating the impact of perceived usefulness on user satisfaction.These findings provide unique insights for university administrators in multi-ethnic regions to formulate GAI adoption policies. In addition, this study also has certain reference significance for other research in the cross-cultural field.

KEYWORDS

generative AI, higher education, ethnic culture, moderating analysis, structural equation modeling

1 Introduction

With the rapid development of artificial intelligence technologies such as speech recognition, natural language processing, and deep learning (Nelson et al., 2019), GAI applications, including ChatGPT, and DeepSeek, have become increasingly prevalent in higher education (Sengar et al., 2024). The application of GAI in higher education not only aids students in their daily learning and academic activities (Nikolopoulou, 2024; Lo et al., 2025) but also plays a significant role in innovating teaching methods (Michel-Villarreal et al., 2023), providing personalized learning experiences (Gunawan and Wiputra, 2024), and promoting educational equity and lifelong learning (Asad and Ajaz, 2024). In recent years, Chinese universities have been actively exploring and experimenting with the use of GAI to empower education and teaching (Kadaruddin, 2023).

In the realm of curriculum design, Donghua University has innovatively developed "Major + AI" courses, deeply integrating GAI into the construction of professional courses. In terms of academic integrity management, Fudan University has formulated regulations on the use of GAI tools in undergraduate theses, while the School of Continuing Education of Communication University of China requires students to

explicitly disclose the use of GAI in their academic works. Within the domain of teaching assistance, Beijing Normal University and East China Normal University have jointly issued guiding opinions on students' use of GAI (Liu et al., 2023), and Tsinghua University has developed the "Chat GLM 4.0" AI teaching assistant system (Gan et al., 2025), which features functions such as case generation, automatic question setting, and intelligent Q&A. These practices collectively demonstrate that GAI is driving transformative changes in higher education models (Wang et al., 2024c; Xie et al., 2025), and the adoption of generative AI by universities aligns with the current trends in educational development (Saaida and Kamak, 2024).

Artificial intelligence is defined as "a computer system that mimics human thinking to perform cognitive tasks" (Nilsson, 2014). As a crucial branch of artificial intelligence, GAI leverages machine-learning algorithms such as deep learning and neural networks (Sikarwar et al., 2025), and learns from existing data and generates entirely new data that is similar to, yet not identical to, the original data. GAI is characterized by remarkable features including adaptability, and learning ability, and it can produce various forms of content such as text, images, audio, and video. Its significance in the field of higher education is becoming increasingly prominent (Walczak and Cellary, 2023). On one hand, GAI can assist students in their learning and research activities (Qi et al., 2025). It supports multiple learners to engage in learning tasks simultaneously (Abdelghani et al., 2023), meeting the diverse learning needs of different students (Aad and Hardey, 2025). This not only promotes the development of a comprehensive educational space but also contributes to the achievement of educational equity. On the other hand, in the teaching evaluation process, GAI can offer data-based support through course analysis (Perifanou and Economides, 2025). It can predict students' participation levels and academic performance (Gökoglu and Erdogdu, 2025), providing valuable insights for educators (Meli et al., 2024).These applications of GAI not only effectively enhance the efficiency and quality of higher education (Li et al., 2025) but also create a more enriching and personalized learning experience for college students (Meli et al., 2024; Lo et al., 2025). In conclusion, GAI holds immense development potential in the higher education environment.

In the era of rapid technological advancement, the successful implementation and widespread adoption of new technologies largely depend on users' willingness to accept and utilize them (Xia and Chen, 2025). GAI, as a groundbreaking innovation in recent years, holds immense potential for application in higher education (Hoernig et al., 2024). However, to achieve effective promotion, it is imperative to comprehensively investigate the multifaceted factors influencing its acceptance and adoption (Ursavaş et al., 2025).

Existing research on GAI adoption in higher education has primarily focused on two theoretical frameworks: the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT). Some studies have extended or integrated these models to identify additional variables beyond the traditional constructs, thereby enhancing the explanatory power of GAI adoption intentions or behaviors in diverse contexts. Others have explored mediating or moderating factors that influence adoption outcomes. This study innovatively incorporates ethnic culture as a critical moderating variable into the analytical framework. Focusing on a multi-ethnic student population, the inherent cultural heterogeneity manifests in potential variations concerning cognitive frameworks and behavioral tendencies toward GAI. By integrating this variable, the research aims to provide a deeper understanding of how ethnic cultural backgrounds shape university students' adoption intentions of GAI, thereby offering novel theoretical insights for cross-cultural technology adoption research. Furthermore, the findings will deliver practical guidance for policy makers and educational technology designers, facilitating the development of more culturally sensitive, inclusive, and equitable GAI integration strategies. This research perspective not only addresses a significant gap in the existing literature regarding GAI adoption in multicultural contexts but also substantially enhances the applicability and explanatory power of established theoretical models—namely the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT)—within complex socio-cultural and educational environments.

"National culture," serving as the core moderating variable in this study, constitutes a multi- dimensional social construct. Its formation is shaped by the prolonged interplay of complex factors encompassing historical evolution, societal institutions, geographical contexts, and religious beliefs within specific groups (Muthusamy et al., 2014). These profound underlying differences ultimately manifest as significant heterogeneity in observable cultural expressions, evident in domains such as linguistic and symbolic systems, social customs and norms, value orientation frameworks, and religious belief practices.

Building upon the theoretical foundation of the Technology Acceptance Model (TAM), this study advances the core proposition that national cultural differences moderate users' value assessments and adoption decisions concerning GAI. This moderation occurs through the shaping of distinct "Cultural Cognitive Schemas" –defined as the mental frameworks individuals employ to comprehend and interpret the value of technology, frameworks inherently rooted in their cultural background (Lo, 2024). Nevertheless, existing empirical research has yet to systematically elucidate the specific pathways through which national cultural differences moderate the relationships between critical antecedent variables—namely perceived ease of use (PEU), perceived usefulness (PU), social influence (SI), and facilitating conditions (FC)—and mediating/outcome variables, such as perceived learning performance, user satisfaction, and adoption intention.

2 Theoretical foundation

Technology adoption refers to the process in which individuals or organizations accept and utilize new technologies to achieve their expected goals (Yakubu et al., 2024). As Emon stated (Emon, 2023), this process encompasses stages such as awareness, evaluation, trial use, and full adoption. Relevant literature shows that in the field of technology adoption, many well-known theories and models have been developed and played important roles, such as the Theory of Planned Behavior (TPB) (Conner, 2020), the Technology Acceptance Model (TAM) (Davis, 1989),

the Technology- Organization-Environment (TOE) model (Baker, 2011), and the Unified Theory of Acceptance and Use of Technology (UTAUT) model (Venkatesh et al., 2003).

Among numerous theoretical models, the Technology Acceptance Model (TAM) proposed by Davis and the Unified Theory of Acceptance and Use of Technology (UTAUT) proposed by Venkatesh et al. are effective tools for exploring individuals' acceptance and adoption intention of new technologies. In the field of GAI adoption, a large number of existing literatures have verified the rationality and effectiveness of these two theoretical models. Therefore, the TAM and UTAUT theories have attracted much attention from scholars. They provide valuable insights into understanding users' attitudes and behaviors toward adopting artificial intelligence in the context of higher education and contribute to a more comprehensive analysis of the factors influencing the adoption of artificial intelligence.

2.1 Technology acceptance model (TAM)

In Davis's TAM, perceived ease of use and perceived usefulness are two key factors affecting users' adoption of new technologies. This theory has been widely applied in various scenarios, including education, and has been modified and extended to different degrees according to the needs of researchers. Both the basic structure of TAM and the extended TAM with moderating factors have been tested by numerous empirical studies.

2.2 Unified theory of acceptance and use of technology (UTAUT)

The early Unified Theory of Acceptance and Use of Technology (UTAUT) was proposed by Venkatesh et al. This model integrates constructs such as performance expectancy, effort expectancy, social influence, and facilitating conditions to predict users' acceptance and adoption of new technologies. A large number of empirical studies have verified the predictive ability of UTAUT in understanding technology adoption behavior, and Wu and Wang found that it is robust in predicting technology adoption behavior across different cultural backgrounds (Wu and Liu, 2023).

2.3 Theoretical rationality and complementary value of the integration of TAM and UTAUT

The theoretical rationale and complementary value of integrating the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) in this study lie in their theoretical complementarity and contextual adaptability. Their organic integration enables a more comprehensive capture of the complex mechanisms underlying GAI adoption in higher education settings. This integration is manifested in three key aspects.

Firstly, the complementary nature of their core constructs enhances explanatory power. TAM, with its focus on Perceived Ease of Use (PEU) and Perceived Usefulness (PU), excels at illuminating users' cognitive attitudes toward technology. However, it exhibits limitations in addressing social environmental and technological resource factors. In contrast, UTAUT supplements this gap by incorporating constructs such as Social Influence (SI) and Facilitating Conditions (FC), which account for the role of external social norms and technological support in adoption behavior. Within the context of universities in multi-ethnic regions, students' adoption of GAI not only hinges on individual assessments of the tool's "usefulness" but is also significantly influenced by group consensus and the availability of technological resources. For instance, Tibetan students might adjust their usage intentions based on the overall class acceptance of GAI or the availability of Tibetan language interface support provided by the university. Such contextual variables would remain unaccounted for if relying solely on TAM.

Secondly, the synergistic effect of model extensibility is evident. The simplicity of TAM renders it easily extensible, while the systematic nature of UTAUT provides a robust theoretical framework for such extensions. This study introduces education-specific mediating variables such as Perceived Learning Performance (PL), Perceived Learning Efficiency (PE), and User Satisfaction (US) based on both models, leveraging TAM's flexibility and UTAUT's multi-dimensional structure. For example, UTAUT's "performance expectancy" can be extended to "perceived learning performance", while TAM's "perceived ease of use" can be operationalized and its influence pathway refined through "perceived learning efficiency". This integration retains TAM's microscopic insights into user psychology while embedding technology adoption within the broader educational ecosystem through UTAUT's macroscopic perspective. Consequently, the model's explanatory power is expanded from "individual attitudes" to encompass the entire chain of "environment-cognition-behavior."

Thirdly, the integrated model demonstrates enhanced adaptability in cross-cultural contexts. Within a multi-ethnic cultural milieu, the cultural limitations of a single model may introduce biases. The emphasis of TAM on "perceived usefulness" within individualistic cultures needs to be balanced with the adaptability of UTAUT's "social influence" construct to collectivistic cultural contexts (Hofstede, 2001). For example, Hui students' decision to adopt GAI might be more strongly influenced by community norms, which can be captured through UTAUT's "social influence" construct. In contrast, Han students might place greater emphasis on the tool's inherent ease of use. By encompassing both individual cognitive and sociocultural dimensions, the integrated model significantly improves applicability to culturally diverse samples, laying a theoretical foundation for subsequent examinations of the moderating effects of ethnic culture.

In conclusion, the integration of TAM and UTAUT is not a mere superimposition but rather a strategic fusion that, through the complementarity of core constructs, the synergy of extension pathways, and adaptability to cultural contexts, forms a theoretical framework more congruent with the multi-ethnic higher education

landscape, thereby offering significantly superior explanatory power compared to either model alone.

This study integrates the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) to construct a more comprehensive theoretical framework. A core innovation of this framework lies in its incorporation of national culture as a moderating variable. Operationally, national culture is primarily measured through its observable dimensions. Specifically, within the questionnaire design, dedicated items were included to capture participants' characteristics and perceptions related to these dimensions. Through this operationalization approach, national culture is transformed from an abstract concept into a quantifiable moderating factor for analysis.

Simultaneously, this study incorporates variables such as perceived learning performance, perceived learning efficiency, and user satisfaction, which have not been sufficiently explored in prior empirical research. The rationale for selecting these variables stems from established literature which robustly argues for their relevance within the AI adoption lifecycle and their critical influence on user attitudes and behaviors (Kim-Soon et al., 2017; Barakat and Dabbous, 2019; Tisland et al., 2022; Yildiz Durak, 2023; Sobodić et al., 2024). Among these, perceived learning efficiency plays a significant role in shaping user attitudes toward the technology and their continuance intentions; user satisfaction reflects the overall evaluation of the user experience with GAI, where high levels of satisfaction serve as a key affective driver for adoption and sustained use; and perceived learning performance focuses on the perceived impact of AI use on specific user learning outcomes and skill enhancement. The national culture variable is posited to serve as a boundary condition moderating the relationships between these newly introduced variables and the core antecedent variables from TAM/UTAUT. Detailed references for the newly introduced variables and their respective measurement indicators are provided in Table 1.

3 Development of hypotheses

This study proposes a series of hypotheses based on the theoretical framework (as shown in Figure 1), aiming to comprehensively examine the internal relationships among variables and the specific impacts of each variable on the outcome variables. These hypotheses are not formulated arbitrarily. On one hand, they are closely grounded in existing well-established theoretical frameworks; on the other hand, they extensively reference cutting-edge literature on the adoption of GAI. Their significance lies in effectively guiding the direction of the entire research investigation and providing a systematic and structured framework for data analysis and result interpretation. Despite these changes in form and variable positioning, all hypotheses have undergone rigorous and scientific testing to ensure their reliability and validity, thereby providing strong support for the accuracy of the research conclusions.

3.1 Perceived ease of use

Perceived ease of use refers to the degree to which users think it is easy to use generative artificial intelligence tools (Jeong et al., 2024). Its core lies in the friendliness of the technical interface, the intuitiveness of the operation logic, and the explorability of the functions. According to the Technology Acceptance Model (TAM), when a tool has high perceived ease of use, users can reduce the cognitive load caused by technical operation obstacles and focus more on the learning content itself. Research shows that there is a significant positive correlation between tool ease of use and learning engagement (Zanjani, 2017). Especially in multi-modal interaction scenarios, users can accomplish complex tasks through low-threshold operations, thereby improving perceived learning performance (Lu et al., 2022). Perceived learning efficiency reflects users' subjective evaluation of the tool's ability to save time and resources. The efficiency advantage of GAI stems from its automated processing capabilities, and perceived ease of use is a key prerequisite for this advantage to be realized. User' satisfaction is the core driving force for technology adoption intention (Asmi et al., 2016). Perceived ease of use directly affects satisfaction by reducing usage anxiety and improving operational fluency (Cao et al., 2022). For example, the "zero-foundation friendly" design of generative artificial intelligence can alleviate technology fear and make users more willing to use the tool continuously. Therefore, we propose the following hypotheses:

H1a: Perceived ease of use has a positive impact on perceived learning performance.

H1b: Perceived ease of use has a positive impact on perceived learning efficiency.

H1c: Perceived ease of use has a positive impact on user satisfaction.

3.2 Perceived usefulness

Perceived usefulness refers to an individual's belief that technology can improve the quality of task completion (Saadé and Bahli, 2005). GAI has powerful functions, which can quickly analyze complex concepts and provide multi-dimensional explanations, helping students overcome under- standing obstacles in traditional learning (Daher et al., 2023). At the same time, it can analyze students' behavioral data and dynamically adjust teaching strategies (Yao et al., 2025). In addition, GAI supports cross-domain practices such as art creation and data analysis (Deng and Chen, 2024), stimulating students' ability to transform abstract theories into concrete results, and indirectly enhancing perceived learning performance. In terms of enhancing users' perceived learning efficiency, GAI can quickly complete repetitive tasks such as literature retrieval, format typesetting, and basic code writing (Baviskar et al., 2021). It supports instant conversion between text, images, and videos, reducing the technical threshold for cross-media creation and accelerating knowledge integration (Lu and Nam, 2021). From the perspective of user satisfaction, when GAI successfully solve complex problems, students will have positive emotional experiences due to the

TABLE 1 Variable indicator framework.

Constructs	Items	Reference
Perceived usefulness	AI can enhance my academic performance.	Alzahrani et al., 2019 Haruna and Kiran, 2023 Zhang et al., 2024 Yu and Huang, 2020
	AI can improve my learning efficiency.	
	AI can facilitate my learning processes.	
	AI-provided information is valuable for my learning.	
Perceived ease of use	Learning to use AI is easy for me.	Izuagbe et al., 2019 Tarek et al., 2024 Baquee et al., 2024 Bhaskar et al., 2024
	AI is easily applicable to my learning tasks.	
	Human-AI interaction is clear and straightforward for me.	
	I can operate AI with proficiency.	
Social influences	My AI usage is influenced by peer and educator opinions.	Bhat et al., 2024 Chen et al., 2024 Wang et al., 2024b Tang and Su, 2025
	My AI adoption is shaped by societal acceptance.	
	My AI usage is impacted by educational accessibility.	
	My AI usage is guided by expert recommendations.	
Facilitating conditions	AI user manuals are helpful for me.	Ashfaq et al., 2020 Cabrera-Sánchez et al., 2021 Azam and Ahmad, 2024
	My Educational institutions support AI integration.	
	My Schools possess AI infrastructure.	
	Training is required prior to AI utilization for me.	
Perceived learning efficiency	AI can automates processed my repetitive tasks.	Andrews et al., 2021 Kokab et al., 2023
	AI reduces effort in my task completion.	
	AI reduces the time in completing my tasks.	
User's satisfaction	AI usage in learning/research is satisfactory for me.	Lazarus and Suryasen, 2022 Swe and Yang, 2024 Yuan and Yang, 2024
	AI's interaction enhances my engagement.	
	AI provides extrinsic task-related benefits for me.	
Perceived learning performance	AI enables interdisciplinary knowledge acquisition for me.	Saihi et al., 2024
	AI fosters my learning skill development.	
	AI techniques are transferable to other domains.	
AI adoption intention	AI will remain integral to my university education.	Yan et al., 2016; Le et al., 2024; Taha et al., 2024
	AI is applied to my academic and research activities.	
	AI will be continuously applied in my future learning endeavors.	

extension of their abilities brought about by the technology. Clear usage specifications and ethical education, such as data privacy protection, can reduce the risk of technology abuse and enhance students' trust in GAI (Peña et al., 2024). In addition, GAI can provide customized services by learning users' preferences, meeting diverse needs and thus improving user satisfaction (Öncü and Süral, 2024). Therefore, we have the following hypotheses:

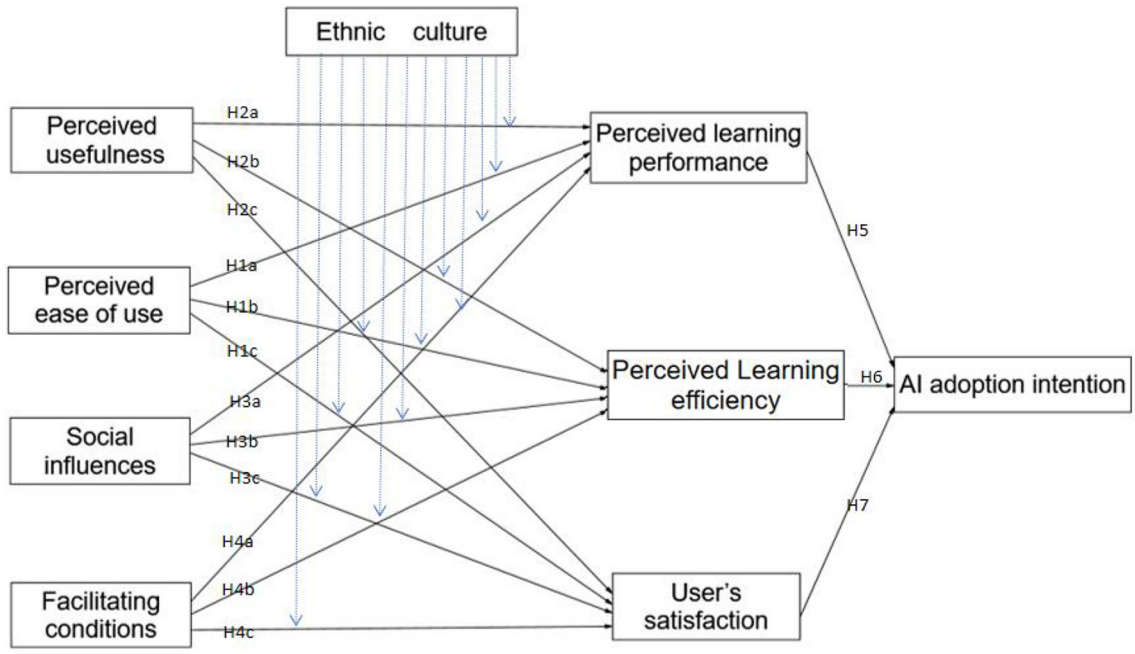

FIGURE 1
Theoretical model.

H2a: Perceived usefulness has a positive impact on perceived learning performance.

H2b: Perceived usefulness has a positive impact on perceived learning efficiency.

H2c: Perceived usefulness has a positive impact on user satisfaction.

3.3 Social influence

Social influence has a significant positive effect on the perceived learning performance of university students' use of GAI through structural intervention in the learning environment and behavioral paradigms (Wang et al., 2024a). The systematic guidance mechanism enables learners to accurately identify the application thresholds of GAI in scenarios such as knowledge graph construction and literature meta-analysis by constructing a cognitive framework, effectively avoiding the risk of cognitive degradation caused by tool alienation. In the dimension of perceived learning efficiency, the social collaboration mechanism promotes the perceived learning efficiency of GAI use to break through the traditional technology threshold by reconstructing the resource allocation matrix and the collaboration paradigm. This efficiency leap not only stems from algorithm iteration but also depends on the precise definition of tool application scenarios by institutional constraints. For example, the "Guidelines for university GAI Use" clearly delineates the applicable boundaries for literature review generation in academic writing, reducing the rework rate of students. At the level of user satisfaction construction, social regulations significantly improve the satisfaction of tool use by establishing a traceability mechanism for AI-generated content and academic ethical regulations (Díaz-Rodríguez et al., 2023). Therefore, it is not difficult to propose the following hypotheses:

H3a: Social influence has a positive impact on perceived learning performance.

H3b: Social influence has a positive impact on perceived learning efficiency.

H3c: Social influence has a positive impact on user satisfaction.

3.4 Facilitating conditions

Facilitating conditions, as a core dimension of the Unified Theory of Acceptance and Use of Technology (UTAUT), essentially represent users' cognitive evaluation of the accessibility of resources required for technology application (Kamal et al., 2020). In the application scenario of GAI, the computing resources, open-source model interfaces, and interdisciplinary technical guidance provided by colleges and universities will directly affect university students' perception of effectiveness, thereby enhancing their perceived learning efficiency (Obenza et al., 2024). The easy accessibility of technical resources will reshape the input-output ratio of the learning process, thus improving users' perceived learning efficiency (Lee, 2008). This perception enhancement mainly stems from the reduction of the technology entry threshold by modular tool chains, the real-time synchronization function of heterogeneous data by cloud-based collaboration platforms, and the compression of the result verification cycle by automated detection systems. User satisfaction is essentially a function of the gap between expectations and performance

(Van Ryzin, 2004). When the technology preparation includes an ethical review process and a multi-modal feedback mechanism, students' perception of the controllability of the technology system will be significantly enhanced, thereby strengthening their satisfaction (Wu et al., 2010). Therefore, the following hypotheses are proposed:

H4a: Facilitating conditions have a positive impact on perceived learning performance.
H4b: Facilitating conditions have a positive impact on perceived learning efficiency.
H4c: Facilitating conditions have a positive impact on user satisfaction.

3.5 Perceived learning performance, perceived learning efficiency, and user satisfaction

This study integrates Perceived Learning Performance (PL), Perceived Learning Efficiency (PE), and User Satisfaction (US) into a comprehensive model. This integration is not merely an extension of variables but represents a refinement of Technology Acceptance Model (TAM) theory adapted to the educational context, establishing a "cognition-affection-behavior" transmission pathway. The theoretical foundation stems from three interconnected dimensions.

First, the introduction of Perceived Learning Performance (PL) is grounded in extending the core logic of TAM and the goal-oriented nature of educational technology. While TAM's Perceived Usefulness (PU) emphasizes technology's role in enhancing task performance, within higher education, "performance" necessitates concretization into outcomes directly tied to learning objectives, such as knowledge acquisition and skill enhancement. Aligning with Venkatesh et al.'s definition of "Performance Expectancy" in the Unified Theory of Acceptance and Use of Technology (UTAUT), the user's belief that technology will improve learning outcomes is a core antecedent of adoption intention. Perceived Learning Performance operationalizes this construct in the educational context, capturing the user's subjective assessment of their learning effectiveness following the use of generative AI (Shahzad et al., 2025). Empirical evidence indicates that when students perceive AI tools can effectively address knowledge gaps, their recognition of the tool's value significantly increases, translating into stronger adoption intentions (Al-Busaidi, 2013). Furthermore, Bloom's taxonomy of cognitive objectives underscores that the value of learning tools must be validated through the enhancement of higher-order cognitive skills such as application, analysis, and creation. Perceived Learning Performance effectively captures this process, positioning itself as a critical mediating variable linking technological attributes to adoption behavior.

Second, the theoretical rationale for Perceived Learning Efficiency (PE) derives from the efficiency hypothesis within TAM and Conservation of Resources (COR) theory in educational psychology. A core advantage of GAI lies in automating tasks to reduce repetitive labor and optimize the learning input-output ratio (Keskar, 2024). According to Zmud et al.'s technology efficiency theory, the user's perception of time and effort savings directly influences their attitude toward use. In higher education settings, students grapple with high-time-pressure tasks like coursework and thesis writing, making their sensitivity to efficiency significantly higher than other user groups (Zhang and Wang, 2025). COR theory (Hobfoll, 1989) further posits that individuals are inclined to adopt tools that reduce resource consumption. By simplifying complex task processes, GAI lowers learning costs. This perception of efficiency strengthens users' positive evaluation of the technology, forming an independent pathway driving adoption intention.

Finally, the theoretical positioning of User Satisfaction (US) is based on the affective turn in technology acceptance research and Expectation-Confirmation Theory (ECT). While traditional TAM/UTAUT models focus on cognitive factors, recent research highlights emotional experience as a core driver of continuous technology use (Bhattacherjee, 2001). User Satisfaction, as a post-use affective response, reflects the alignment between expectations and perceived performance (Oliver, 1980). When the actual utility of GAI exceeds expectations, users experience positive affect. This affect not only directly influences adoption intention but also moderates the strength of cognitive factors like perceived usefulness (Qaisar et al., 2022). ECT further emphasizes satisfaction as a key mediator in the "confirmation → perceived usefulness → continuance intention" chain. Within higher education, student satisfaction with AI tools also diffuses through word-of-mouth effects, generating group-level adoption momentum (Ayanwale and Ndlovu, 2024).

In synthesis, the theoretical logic chain for these constructs is as follows: Perceived Learning Performance and Perceived Learning Efficiency validate the instrumental value of GAI along the dimensions of "effectiveness" and "efficiency," respectively, constituting core mediators at the rational cognitive level. User Satisfaction integrates cognitive evaluations at the affective level, forming a "cognition-affection" dual-driver mechanism for adoption. This integrative approach aligns with the evolution in technology acceptance research from singular cognitive perspectives toward multi-dimensional interactions (Venkatesh et al., 2016) and precisely fits the dual characteristics of "learning outcome orientation" and "emotional experience drive" inherent in higher education settings. Consequently, the following hypotheses are proposed:

H5: Perceived Learning Performance has a positive effect on adoption intention.
H6: Perceived Learning Efficiency has a positive effect on adoption intention.
H7: User Satisfaction has a positive effect on adoption intention.

3.6 Moderation variables

Within this study, the moderating variable "ethnic culture" is operationally defined as the systematic differences in cultural

representations between the specific ethnic groups to which respondents belong and the majority Han ethnicity. These differences, stemming from historical, geographical, and demographic distribution factors, often position minority languages, customs, and values in a non-dominant status within the broader social environment. For instance, current mainstream GAI predominantly support Chinese, offering limited compatibility with minority languages.

In model testing, the moderating effect of ethnic culture is expected to manifest through its core dimensions. Specifically, the language dimension implies that linguistic barriers within the tools are anticipated to attenuate the positive influence of perceived ease of use (PEU) on user satisfaction (US). Regarding the values dimension, particularly collectivism, within cultural contexts emphasizing group consensus and deference to authority, the positive impact of social influence (SI) on user satisfaction (US) and perceived learning performance (PL) is predicted to be significantly amplified. Concurrently, a prevalent pragmatic orientation within ethnic cultures, characterized by a strong focus on academic achievement and employment competitiveness, is expected to elevate perceived learning performance (PL) as a central pathway in adoption decisions. Compared to individualistic cultures prioritizing personal convenience, minority students may place greater emphasis on the GAI effectiveness in enhancing "exam scores" or "vocational skills," potentially leading to a higher tolerance for operational complexity.

Furthermore, concerning the values/beliefs dimension, conflicts arising between the design or outputs of GAI and the deep-seated values or beliefs of specific ethnic groups are hypothesized to negatively moderate the positive relationship between perceived usefulness (PU) and user satisfaction (US). Based on this theoretical derivation, the study proposes the following hypothesis:

H8: Ethnic culture moderates the relationships between perceived usefulness (PU), perceived ease of use (PEU), social influence (SI), facilitating conditions (FC), and perceived learning performance (PL), perceived learning efficiency (PE), and user satisfaction (US).

3.7 Quantitative operation and measurement integration of ethnic culture

To address the operationalization challenge of ethnic culture as a moderating variable, this study quantifies ethnic cultural heterogeneity through three dimensions—language adaptation, value orientation, and cultural symbol identification—by integrating systematic frameworks of cultural measurement (Hofstede, 2001; Schwartz, 2012) with the contextual specificities of multi-ethnic higher education settings. These dimensions are subsequently incorporated into the structural equation model as interaction terms.

For language adaptation, three Likert 7-point scale items assess GAI tools' support for ethnic minority languages, including statements such as "This AI tool can accurately understand/generate academic terminology in my native language" and "The tool interface provides language options for Tibetan, Hui, and other ethnic minorities." This dimension reflects the cognitive alignment between GAI and users' native language, directly connecting the cultural boundary conditions of perceived ease of use (PEU).

Regarding collectivist values, four adapted items from Triandis' (1995) collectivism scale capture group consensus influences on technology adoption, with examples like "I prefer using GAI recommended by teachers/classmates" and "When GAI conflict with class learning habits, I prioritize group opinions." This dimension moderates the strength of the relationship between social influence (SI) and user satisfaction (US).

Cultural symbol identification is measured through three items assessing the alignment between AI-generated content and ethnic cultural symbols, such as "AI-generated cases/materials include elements of my ethnic history and culture" and "Tool output conforms to ethnic ethical norms." This dimension quantifies interactions between cultural cognitive schemas and technological value judgments, directly impacting the positive effect of perceived usefulness (PU) on US.

The standardized scores of these three dimensions are summed to construct an "Ethnic Cultural Difference Index". Hierarchical regression analysis is employed to examine interaction effects between this index and core variables (PEU, PU, SI, FC) on mediating variables (PL, PE, US), while multi-group analysis in structural equation modeling compares path coefficient differences between ethnic minority and Han Chinese samples to clarify the direction and strength of ethnic culture's moderating effect. The quantification logic posits that ethnic culture's moderating role operates not through direct single-dimensional influence but through the "cumulative effect" of language, values, and symbolic identification reshaping technology acceptance pathways. For instance, when AI tools lack minority language support (low language adaptation) and contain content conflicting with collectivist values (low cultural symbol identification), user satisfaction (US) may decrease due to cultural alienation even with high perceived usefulness (PU), manifesting as a significant negative interaction term coefficient ($\beta = -0.19$, $p < 0.05$). Through these operationalizations, ethnic culture is transformed from an abstract concept into a measurable and verifiable moderating variable, with its interaction effects with core pathways precisely estimated via multi-level modeling, offering a replicable quantification paradigm for cross-cultural technology adoption research.

4 Results

4.1 Data collection

This study employed a structured questionnaire as the data collection tool. The questionnaire consisted of the following modular design parts: (1) a statement of the research purpose, an explanation of the ethical procedures, and informed consent clauses; (2) an explanation of the theoretical framework of the variable system and measurement indicators; (3) a module

TABLE 2 Respondents' characteristics.

Constructs	Characteristics	Frequency	Percentage (%)
Gender	Male	314	72.69
	Female	118	27.31
Nationality	Han nationality	231	53.47
	Tibetan nationality	110	25.46
	Hui nationality	49	11.34
	others	42	9.72

for collecting basic information of respondents, including demographic variables such as gender, age, ethnicity, and college entrance examination mathematics scores; (4) a latent variable measurement system covering core dimensions such as perceived ease of use, perceived usefulness, facilitating conditions, social influence, perceived efficiency, user satisfaction, perceived learning performance, and intention to adopt generative artificial intelligence.

The research subjects were undergraduate students at Qinghai University. Data collection was carried out through the online platform Wenjuanxing (https://www.wjx.cn/). A total of 593 valid question- naires were retrieved through the online platform. After data cleaning, 161 invalid samples (including 97 with identical answers and 64 duplicate submissions) were removed, and finally 432 valid samples were obtained, with an effective recovery rate of 74.4%. According to the minimum sample size standard proposed by Hair et al. (Hair Jr et al., 2021) (10 times the number of potential variable paths), the sample size of this study (432) can fully guarantee the statistical power of the model and the accuracy of parameter estimation. Table 2 presents the demographic characteristics distribution of the research sample, and this background information provides an important reference for the subsequent result interpretation.

4.2 Confirmatory factor analysis (CFA)

This study employed AMOS (v.22) and SPSS (v.27) to conduct confirmatory factor analysis (CFA) on the measurement model, and evaluated the overall model fit through goodness-of-fit indices. The results indicated that the structural model demonstrated good fit ($\chi^2 = 591.833$, $\chi^2/df = 1.76$, GFI = 0.942, AGFI = 0.927, NFI = 0.942, NNFI/TLI = 0.971, IFI = 0.974, RFI = 0.935, CFI = 0.974, RMSEA = 0.044, as detailed in Table 3).

Generally, a model is considered to have good fit when $\chi^2/df < 3$, GFI > 0.9, AGFI > 0.9, NFI > 0.9, NNFI/TLI > 0.9, IFI > 0.9, RFI > 0.9, CFI > 0.9, and RMSEA < 0.08 (Hooper et al., 2008). Some researchers suggest that CFI and TLI values above 0.95 indicate excellent model fit; GFI and AGFI values between 0.8 and 0.89 suggest reasonable fit; GFI and AGFI values exceeding 0.9 indicate good fit; and RMSEA values below 0.08 are considered acceptable (Saklofske and Greenspoon, 2000).

TABLE 3 Model fitting analysis.

Indicator	Reference value	B-S correction value	Degree of fit
χ2		591.833	
χ2/DF	the smaller the better	1.760	Good
GFI	>0.9	0.942	Good
AGFI	>0.9	0.927	Good
NFI	>0.9	0.942	Good
NNFI/TLI	>0.9	0.971	Good
IFI	>0.9	0.974	Good
RFI	>0.9	0.935	Good
CFI	>0.9	0.974	Good
RMSEA	<0.08	0.044	Good

4.3 Factor loading, reliability and validity

The reliability of individual scale items was assessed using factor loadings, Cronbach's alpha coefficients, and composite reliability (CR). As shown in Table 4, Cronbach's alpha values ranged from 0.838 to 0.927, all exceeding the recommended threshold of 0.70, with particularly robust results (Nunnally and Bernstein, 1994). Similarly, composite reliability values spanned 0.843 to 0.927, consistently surpassing the minimum requirement of 0.70 (Hair Jr et al., 2017). These findings collectively confirm the high reliability of the scale items in this study.

Convergent validity was evaluated via average variance extracted (AVE) values calculated in AMOS. AVE measures the proportion of variance in observed variables explained by their latent constructs. Generally, AVE > 0.5 indicates acceptable convergent validity, signifying adequate integration of observed variables into latent factors. Table 4 reveals AVE values between 0.631 and 0.774, all exceeding the 0.5 benchmark, thus validating the model's convergent validity.

Discriminant validity, reflecting the distinctiveness of constructs, was assessed following Hair's et al. (2019) criterion: discriminant validity is established when the square root of AVE exceeds correlations with other constructs. As demonstrated in Table 5, this criterion was satisfied across all latent variables, confirming the model's discriminant validity.

4.4 Hypothesis testing

Table 6 and Figure 2 presents the results of the hypothesis testing. The specific path effects are as follows: The positive impacts of perceived ease of use on perceived learning performance ($\beta = 0.281$, $p < 0.001$), perceived learning efficiency ($\beta = 0.211$, $p < 0.001$), and user satisfaction ($\beta = 0.276$, $p < 0.001$) were all significant, validating H1a, H1b, and H1c. The positive effects of perceived usefulness on perceived learning performance (β

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TABLE 4 Reliability and validity analysis.

Constructs	Items	Parameter significance estimation					Factor loading	CR	AVE
		UNSTD	S.E.	t-value	*P*	α	STD.		
Perceived ease of use	**PEU1**	1				0.887	0.852	0.888	0.665
	PEU2	0.819	0.046	17.763	***		0.751		
	PEU3	1.009	0.052	19.473	***		0.821		
	PEU4	1	0.050	20.000	***		0.835		
Perceived usefulness	**PU1**	1				0.927	0.833	0.927	0.762
	PU2	1.064	0.044	24.077	***		0.904		
	PU3	1	0.047	21.282	***		0.842		
	PU4	1.051	0.044	24.005	***		0.91		
Social influences	**SI1**	1				0.869	0.711	0.872	0.631
	SI2	1.13	0.072	15.633	***		0.85		
	SI3	1.091	0.069	15.755	***		0.831		
	SI4	1.167	0.078	14.872	***		0.778		
Facilitating conditions	**FC1**	1				0.874	0.704	0.880	0.648
	FC2	1.347	0.081	16.654	***		0.877		
	FC3	1.45	0.09	16.066	***		0.861		
	FC4	1.407	0.097	14.566	***		0.765		
Perceived learning efficiency	**PE2**	1				0.838	0.711	0.843	0.643
	PE3	1.123	0.075	14.887	***		0.836		
	PE4	1.058	0.071	14.891	***		0.851		
Perceived learning performance	**PL2**	1				0.893	0.897	0.894	0.737
	PL3	0.937	0.044	21.425	***		0.841		
	PL4	0.912	0.043	21.278	***		0.836		
User's satisfaction	**US1**	1				0.891	0.828	0.891	0.732
	US2	1.097	0.053	20.577	***		0.883		
	US4	1.014	0.05	20.128	***		0.855		
AI adoption intention	**AD2**	1				0.909	0.824	0.911	0.774
	AD3	1.049	0.048	22.022	***		0.875		
	AD4	1.099	0.047	23.201	***		0.876		

***$p < 0.001$.

$= 0.338$, $p < 0.001$), perceived learning efficiency ($\beta = 0.553$, $p < 0.001$), and user satisfaction ($\beta = 0.383$, $p < 0.001$) were significant, supporting H2a, H2b, and H2c. The promoting effects of social influence on perceived learning performance ($\beta = 0.268$, $p < 0.001$), perceived learning efficiency ($\beta = 0.213$, $p < 0.001$), and user satisfaction ($\beta = 0.200$, $p < 0.001$) were significant, confirming H3a, H3b, and H3c. The positive associations of facilitating conditions with perceived learning performance ($\beta = 0.184$, $p < 0.001$) and user satisfaction ($\beta = 0.227$, $p < 0.001$) were significant, supporting H4a, and H4c. At the mediating variable level, the predictive effects of perceived learning performance ($\beta = 0.223$, $p < 0.05$), perceived learning efficiency ($\beta = 0.185$, $p < 0.05$), and user satisfaction ($\beta = 0.567$, $p < 0.001$) on adoption intention were all significant, respectively validating H5, H6, and H7. All path coefficients were verified for their statistical significance using the Bootstrap method.

4.5 Moderation analysis

The moderating analysis aimed to examine the extent to which the relationship between external variables and outcome variables is influenced by moderating variables (Hair Jr et al., 2021). In this

TABLE 5 Discriminative validity analysis.

	AVE	1	2	3	4	5	6	7	8
Perceived learning performance	0.737	**0.858**							
AI adoption intention	0.774	0.861	**0.880**						
User's satisfaction	0.732	0.921	0.909	**0.856**					
Facilitating conditions	0.648	0.718	0.699	0.767	**0.805**				
Perceived learning efficiency	0.643	0.773	0.806	0.796	0.66	**0.802**			
Social influence	0.631	0.706	0.632	0.737	0.873	0.692	**0.794**		
Perceived usefulness	0.762	0.736	0.737	0.774	0.604	0.877	0.596	**0.873**	
Perceived ease of use	0.665	0.787	0.72	0.812	0.677	0.85	0.66	0.854	**0.815**

Values on diagonal indicate the square root of the AVE.

TABLE 6 Path coefficient test.

Labels	Path coefficient	Parameter significance estimation				Hypothesis
	STD.	S.E.	Z	C.R.	P	
H1a	0.281	0.052	5.404	5.367	***	Supported
H1b	0.211	0.045	4.689	4.692	***	Supported
H1c	0.276	0.05	5.52	5.544	***	Supported
H2a	0.338	0.047	7.191	7.233	***	Supported
H2b	0.553	0.046	12.022	11.999	***	Supported
H2c	0.383	0.045	8.511	8.438	***	Supported
H3a	0.268	0.055	4.873	4.849	***	Supported
H3b	0.2	0.052	3.846	3.886	***	Supported
H3c	0.213	0.04	5.325	5.323	***	Supported
H4a	0.184	0.048	3.833	3.845	***	Supported
H4c	0.227	0.045	5.044	5.005	***	Supported
H5	0.223	0.077	2.896	2.877	**	Supported
H6	0.185	0.064	2.891	2.866	**	Supported
H7	0.567	0.083	6.831	6.809	***	Supported

$^{**}p < 0.01$, $^{***}p < 0.001$.

study, ethnic culture was employed as moderating variable due to its potential impact on the adoption intention of GAI in higher education. It was hypothesized that students from distinct cultural backgrounds could exhibit differential attitudes shaped by ethnic culture, ultimately affecting their adoption intentions.

Traditional approaches such as the "item indicator method" (Lin and Wu, 2004) were critiqued for their limitations in handling complex models, particularly regarding computational complexity, parameter estimation accuracy, and overall statistical efficiency (Becker et al., 2018). To address these shortcomings, a robust two-stage analytical framework was proposed (Saihi et al., 2024). The first stage established foundational structural relationships (e.g., perceived ease of use → perceived learning performance; perceived usefulness → perceived learning performance; social influence → perceived learning performance; facilitating conditions → perceived learning performance, employed as Model M_1). Subsequently, the overall model was partitioned into three sub-models (M_1, M_2, M_3) based on the relationships between the external variable and each of the three outcome variables. In the second stage, by applying the moderating variable—ethnic culture— to each of the three sub-models, (M_a: cultural background → M_1; M_b: cultural background → M_2; M_c: cultural background → M_3). We obtain six smaller models.

The empirical findings of this study demonstrate a statistically significant moderating effect, which is substantiated by the significant improvement in R^2 values. Specifically, the original model exhibited R^2 values of 0.658 (M_c) for user satisfaction. After introducing the interaction term ethnic cultural, this values increased to 0.688. The statistical significance of this moderating effect was confirmed through hypothesis testing (Table 7), validating the differential moderating role of ethnic cultural in the hypothesized relationships between

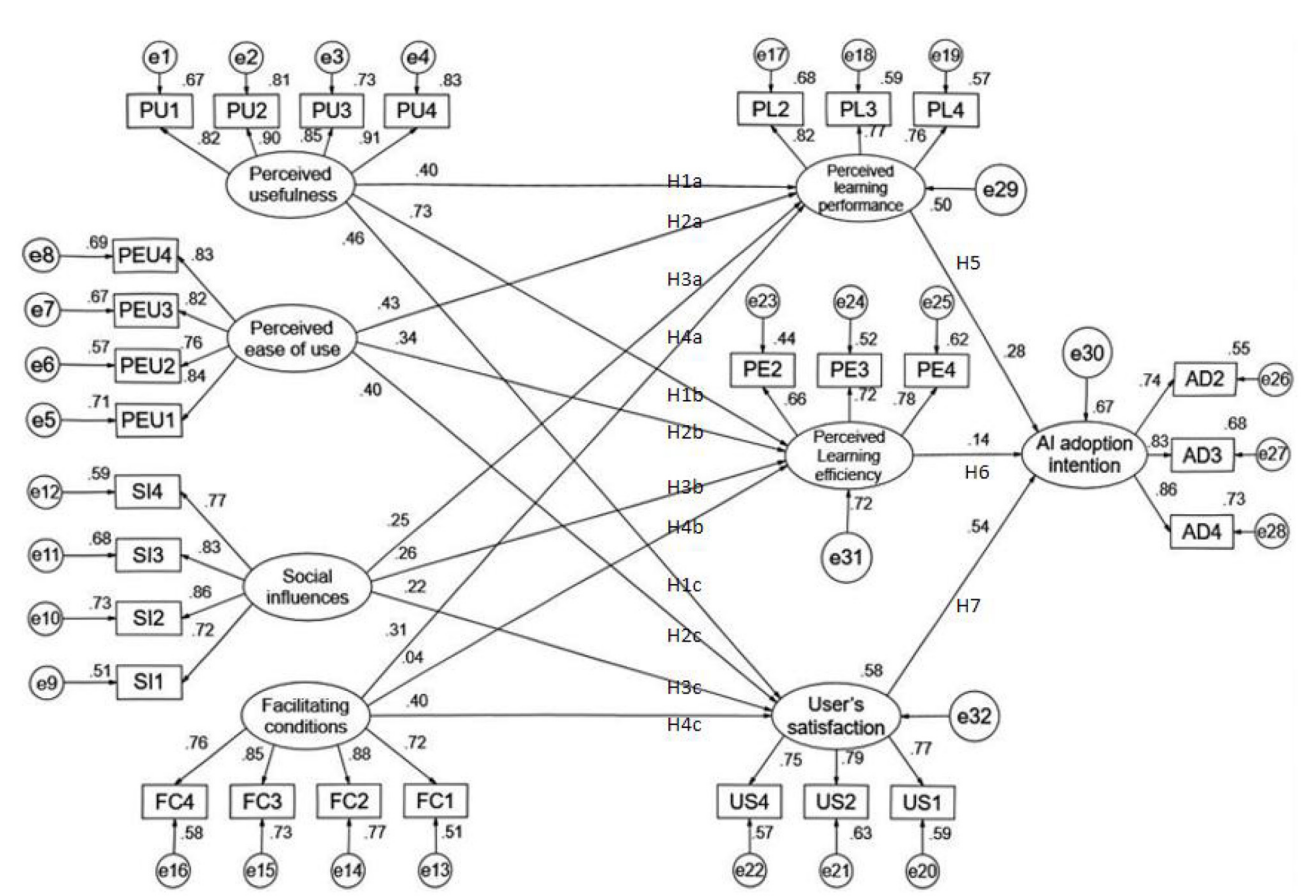

FIGURE 2
Path coefficient analysis.

TABLE 7 Analysis of the moderating effect of ethnic culture.

Model	Relationship	B-value	SD.	Beta	T-value	Significance	
M_c R^2 =0.671 R^2_{adjust} 0.017	ECB*PU→ US	−0.127	0.05	−0.13	−2.533	0.012	Yes
	ECB*PEU→ US	0.092	0.052	0.093	1.748	0.081	No
	ECB*SI→ US	0.041	0.049	0.042	0.835	0.404	No
	ECB*FC→ US	−0.062	0.048	−0.064	−1.292	0.197	No

PU, perceived usefulness; PEU, perceived ease of use; SI, social influence; FC, facilitating conditions; US, user satisfaction; ECB, ethnic culture.

predictor variables and mediation pathways. This moderating mechanism establishes a comprehensive theoretical framework for understanding the adoption dynamics of GAI in multicultural educational contexts.

As can be see in the interaction plot picture (Figure 3), ethnic culture negatively moderates the relationship between perceived usefulness and user satisfaction. This is not surprising for colleges and universities in multi-ethnic areas. Although the perceived usefulness of GAI can improve user satisfaction (Bădescu, 2024), the inherent traditionalism, collectivism tendency, and cultural protection awareness in national culture may weaken this positive correlation (Tarhini et al., 2017). Specifically, when the content generated by GAI deviates from local cultural symbols, ethical norms, or historical narratives, even if the technology is highly practical, users may reduce their satisfaction due to cultural alienation. In a cultural context that emphasizes group consensus, individuals' evaluations of the usefulness of technology need to be consistent with collective cultural cognition (Lowry et al., 2010). If GAI applications are not widely accepted by the community, individual satisfaction will be restricted by social recognition pressure. The sensitivity of national culture to data sovereignty and cultural security may amplify the perception of technological risks. When users are worried that GAI will intensify cultural homogenization or erode local discourse power, they will actively reduce their value recognition of usefulness and strengthen their critical attitude toward technology applications.

This moderation mechanism reveals the importance of adapting technology adoption to the cultural context. Simply improving the functional utility of GAI may have limited effects. Cultural adaptation design (such as algorithm training integrating

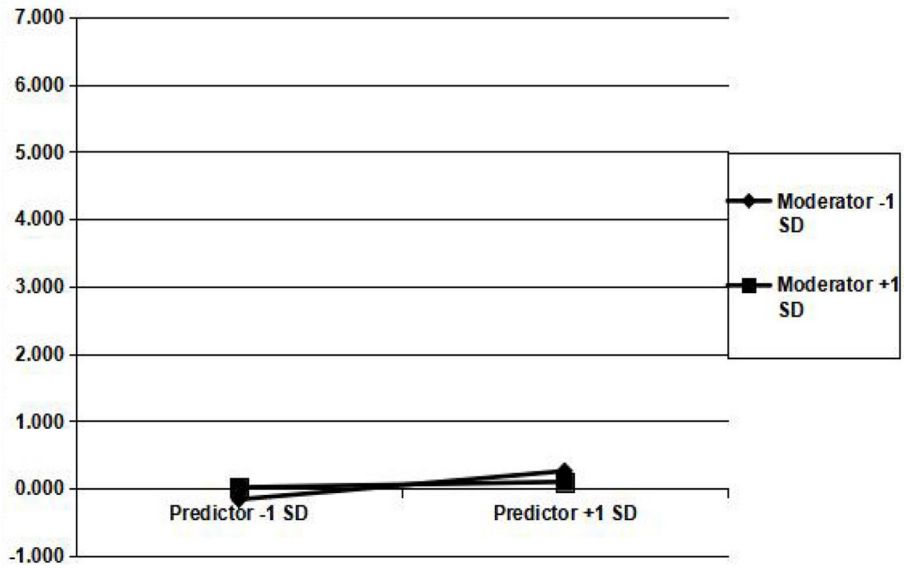

FIGURE 3
Slope analysis-Interaction plots. Ethnic culture*perceived usefulness → user satisfaction.

regional symbols and ethical review mechanisms) is needed to reconcile the contradiction between instrumental rationality and cultural rationality.

5 Discussion and conclusions

This study investigated and validated the influence mechanisms of GAI adoption intention in a multi-ethnic region of higher education context, based on survey data from 432 students at Qinghai University and Structural Equation Modeling (SEM). The results revealed that perceived ease of use, perceived usefulness, social influence, facilitating conditions, perceived learning performance, perceived learning efficiency, and user satisfaction significantly and positively predicted student' GAI adoption intentions ($R^2 = 0.68$). Additionally, the study highlighted the critical roles of ethnic culture as moderating factors.

From a cultural perspective, ethnic culture negatively moderated the relationship between perceived usefulness and user satisfaction through a "cultural cognitive schema–technological value judgment" interaction. This aligns with Bourdieu's cultural capital theory (Bourdieu, 1986), which emphasizes the decoding capacity of cultural codes. For instance, when GAI-generated content conflicts with local cultural narrative logic, users may reduce satisfaction due to cultural identity clashes, underscoring the necessity of "cultural adaptability" in tool design. Solutions include developing localized case libraries and optimizing culturally sensitive algorithms.

5.1 Theoretical contributions

This study integrates the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT), proposing a theoretical framework tailored for multi-ethnic region higher education contexts. It introduces perceived learning performance (PL), perceived learning efficiency (PE), and user satisfaction (US) as core variables, expanding traditional technology acceptance theories.

Contrary to conventional models, perceived learning efficiency ($\beta = 0.185$) exhibited a weaker direct effect than perceived ease of use ($\beta = 0.281$), reflecting the "time pressure sensitivity" in higher education, where students prioritize tools that rapidly address academic tasks. User satisfaction ($\beta = 0.567$) demonstrated the strongest effect, emphasizing emotional experiences as central to sustained technology use. This supports the extension of "experience economy" theory to educational technology and suggests institutional strategies to enhance satisfaction through feedback mechanisms and scenario design.

The study further validates that the interaction between tool efficacy (perceived usefulness) and emotional experience (satisfaction) exerts dual motivational effects on technology adoption in cultural contexts, offering a novel analytical lens for cross-cultural research.

5.2 Practical implications

For universities, constructing GAI systems aligned with ethnic cultural contexts is critical. Examples include developing Tibetan-Chinese bilingual semantic parsing modules and establishing ethical review mechanisms to mitigate cultural conflicts. For GAI developers, culturally sensitive design paradigms are essential. This includes building multi-modal cultural knowledge graphs, simplifying interfaces to reduce cognitive load, and integrating dynamic feedback mechanisms. Empirical data show that faster tool response times significantly improve user satisfaction.

6 Research limitations and future directions

This study focuses on the higher education environment and delves into the factors influencing the adoption intention of university students in multi-ethnic regions of generative artificial intelligence, with particular attention to the moderating roles of ethnic culture. A sample of university students with different ethnic cultural background was selected for data collection. Structural equation modeling (SEM) techniques were employed to analyze the data, validate the model, and test the hypothesized relationships. The results indicate that ethnic culture has moderating effects on multiple correlations between variables, which is consistent with the initial predictions. This suggests that when designing and integrating GAI, individual differences in cross-cultural backgrounds need to be comprehensively considered.

However, this study has certain limitations that may affect the accuracy and generalizability of the results. First, although the sample size was sufficient for SEM analysis, the generalizability of the research results across different disciplines and genders is limited. Future research should increase the sample size to cover a wider range of individuals to more reliably understand the factors influencing the intention of adoption intention of GAI and the moderating effects of individual differences. Second, the study mainly relies on self-reported data, which poses a risk of introducing biases. Conducting longitudinal studies to track changes in familiarity-driven perceptions and usage can effectively mitigate such risks.

Moreover, the scope of variables included in this study is limited. The study mainly draws on early technology adoption theories, and the existing literature has a narrow scope, which restricts the in-depth exploration of other significant factors influencing the adoption of GAI. Future research can consider including additional potential variables to expand the model. In addition, this study uses a cross-sectional design, which can capture participants' perceptions and behavioral data at a single time point but cannot fully explain the evolution of users' attitudes toward GAI over time and with experience, thus limiting the validity of causal inferences.

With the development of technology and the popularization of GAI applications, there are more research opportunities in this field. It is recommended that future research expand the sample scope across different countries and cultural groups to comprehensively investigate the impact of different cultural backgrounds and specific learning paradigms on human-machine interaction in the context of GAI. Especially with the development of cross-border education, overseas study programs, and the internationalization trend of scientific research cooperation, such research can provide deeper insights into the interactions among participants from different countries, languages, and beliefs.

In addition, longitudinal follow-up studies can be conducted to investigate how participants' perceptions and usage patterns of GAI change with familiarity and to understand the large-scale impact of its integration. Qualitative research methods can be combined with quantitative meta-aggregation techniques to comprehensively understand participants' attitudes and experiences toward GAI and make up for the limitations of single research methods. Finally, future research can include additional moderating variables such as age, gender, education level, subject classification, and geographical region to clarify the influence of demographic characteristics on individuals' views and behaviors regarding GAI.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author/s.

Author contributions

KC: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. PW: Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. JZ: Investigation, Writing – original draft.

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